# Zhixin (Liam) Zeng

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## Education

**Northeastern University** — Vancouver, BC, Canada
*M.S. in Computer Science; GPA: 3.9/4.0* — Sep. 2023 – Dec. 2024

**Sichuan University** — Chengdu, China
*B.S. in Cybersecurity; GPA: 3.8/4.0* — Sep. 2019 – Jun. 2023

## Publications and Preprints

[1] **Zhixin Zeng** and Jiahao Zhong. Cassette-Editor: Shared Timeline Semantics for Human and Agent Editing. *Preprint*, arXiv:2610.37690, 2026. Latest version on Zenodo.

[2] Wenbin Zhao, Peisong He, **Zhixin Zeng**, and Xiong Xu. Fake News Detection Based on Knowledge-Guided Semantic Analysis. *Electronics*, 13(2):259, 2024.

[3] Yaxuan Wang, **Zhixin Zeng**, and Qijun Zhao. Evaluating the Perceived Safety of Urban City via Maximum Entropy Deep Inverse Reinforcement Learning. In *Proceedings of the Asian Conference on Machine Learning (ACML 2022)*, PMLR 189:1085–1100, 2023.

[4] Yaxuan Wang*, **Zhixin Zeng***, Qiushan Li, and Yingrui Deng. A Complete Reinforcement-Learning-Based Framework for Urban-Safety Perception. *ISPRS International Journal of Geo-Information*, 11(9):465, 2022. (*equal contribution)

## Research Experience

**Cassette-Editor: Shared Timeline Semantics for Human and Agent Editing** — Jan. 2025 – Present
*Lead author and system designer; co-founder of Cassette* — Paper [1]

*A video-editing agent must keep dependent timeline state consistent while the project stays open to manual revision. Cassette studies how to divide that work: the model chooses edits; the editor derives their consequences, enforces project invariants, and locates source evidence in the current cut.*

- Gave the user and the agent **shared write semantics**: a manual gesture and an agent patch compile into the same typed document write, which one executor applies, deriving its consequences for later clips, titles, and transitions; one rulebook judges the result and one undo history records it, so users revise agent edits like their own. A write commits atomically or changes nothing.
- Made **the timeline document the action language**, with **editing parity by construction**: instead of a catalog of per-operation tools, the agent writes typed changes derived mechanically from the document schema, each translating to one canonical change and back, so it can express any content change a manual gesture makes.
- Built **the editor as a compiler** for agent edits: a front end repairs unambiguous slips and reports what it left out; the rulebook applies a fix that is unique and keeps the intent, and rejects other violations at their entity and field with typed repair options and the reads that would supply missing facts.
- Built **evidence that follows the edit**: shot descriptions, word-timed speech, sound events, and beats, analyzed once in source time, reach edited frames through retiming, reversal, and nested sequences using the mapping code shared by the rulebook, preview, and renderer.
- **Evaluation** (31 authored requests, released GPT-6 Luna): the full harness passed in **92/93** trials against 48/93 with basic tools, at a third of the agent time per pass. Follow-ups isolate components: removing derived consequences or rule enforcement cut passes (230/232→199/232; 46/48→10/48); edit-aware reads and full diagnostics cut agent time and repair tokens. On **TVSum**, it agreed with human annotators more than a general coding agent (Codex CLI) on the same model, **F1 23.8 vs. 20.1**.
- Built and deployed Cassette (app.trycassette.online) as primary developer: browser editor, macOS app with local GPU export, agent runtime, cloud rendering, MCP service, Final Cut Pro exchange. Released **oh-my-cassette**, an open-source plugin connecting coding agents to the editor.

**Knowledge-Guided Fake News Detection** Oct. 2022 – May 2023
*Undergraduate Researcher, Sichuan University; B.S. thesis* Paper [2]
Advisor: Prof. Peisong He

*Investigated how agreement between news claims and external knowledge can improve fake news detection and generalization beyond the training domain.*

- Extracted subject–relation–object triplets with REBEL and proposed **fuzzy-matching alignment** to the CommonSense Knowledge Graph, accommodating variations in entity and relation wording without manual labeling or entity linking.
- Independently developed a **knowledge-guided dual-branch model**: aggregated TransE triplet embeddings to guide a BERT text branch, scored claim plausibility in knowledge-embedding space, and fused the branches with learned interaction weights.
- Implemented the full pipeline and conducted all experiments. Training on LUN and testing on the 360-article SLN set yielded **94.45% accuracy and 94.38% macro-F1**, compared with 89.12% and 89.29% for SMHN. On a separate 200-document timing benchmark, average processing time was 0.325 s per document versus 1.257 s for SMHN.
- Established the contribution of external knowledge through ablations: removing the knowledge branch reduced accuracy to **86.28%**; separate ablations confirmed the contributions of the aggregation and scoring modules.

**Urban Safety Perception as Human-Like Decision Making** Oct. 2021 – Aug. 2022
*Undergraduate Researcher, Sichuan University* Papers [3, 4]
Advisors: Prof. Qijun Zhao, Dr. Qiushan Li

*Modeled perceived urban safety as a sequential decision problem, using crowdsourced judgments to learn interpretable visual preferences from a small dataset.*

- Built the **SmallCity dataset**: selected 522 geographically balanced street views from 45,412 Baidu images, stitched four camera headings, and collected balanced pairwise safety judgments through a crowdsourcing website. Converted judgments to scores using TrueSkill and LambdaRank.
- Designed an **interpretable scene representation** with SegFormer after comparing four segmentation models, encoding greenery, sky and building fields of view, wire poles, car count, and visual entropy.
- Formulated safety judgment as an **MDP** in a custom Gym environment and trained PPO, A2C, SAC, and D3QN with Ray RLlib. With an expert-designed reward, D3QN achieved **0.686 AUC**, compared with 0.617 for SVM and 0.550 for a five-layer CNN [4].
- Extended the framework with **maximum-entropy deep inverse reinforcement learning** to recover rewards from crowd judgments. IRL+D3QN achieved **69.3% F1 and 0.709 AUC**, compared with 66.7% and 0.618 for ViT; ablations identified greenery as the strongest factor [3].